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CUSTOMER SUPPORT TICKET ANALYSIS REPORT

An Analysis of Customer Complaints, Support Performance, and Resolution Efficiency

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Project Overview

This project analyzes customer support ticket data to evaluate service performance, identify recurring issues, and uncover patterns affecting customer satisfaction.

Customer support plays a critical role in retaining customers and ensuring a positive user experience. This analysis aims to provide actionable insights that will improve operational efficiency, reduce repeated complaints, and enhance overall service delivery.

Project Objectives

The objectives of this analysis are to:

- Analyze customer complaint patterns
- Identify the most frequent issue types
- Evaluate support agent performance
- Assess resolution efficiency
- Identify high-risk customers
- Provide actionable, data-driven recommendations.

Data Cleaning & Preparation

To ensure data accuracy and consistency, the following steps were performed:

- Removed duplicate records to avoid double counting
- Standardized text values using Find & Replace (e.g., “delay” → “Service Delay”)
- Trimmed extra spaces to ensure consistency
- Formatted date columns correctly
- Handled missing values:

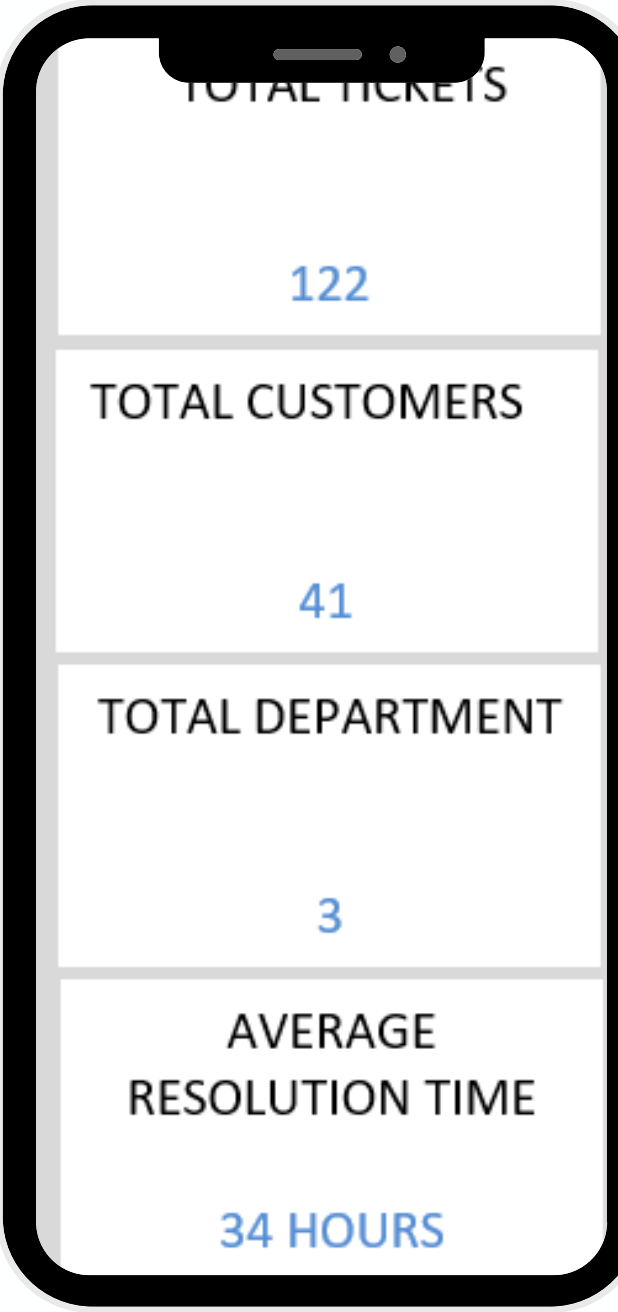
  Issue Type → “Unknown Issue”

  Priority → “Unspecified”

These steps ensured the dataset was clean, structured, and suitable for analysis.

Key Performance Indicators (KPIs)

- Total Tickets: 122
- Total Customers: 41
- Total Departments: 3
- Average Resolution Time: 34 hours



These KPIs provide a high-level overview of customer support operations.

Ticket Status Analysis

The distribution of tickets by status is as follows:

- Resolved: 66 tickets
- Open: 34 tickets
- Closed: 22 tickets

A majority of tickets have been resolved, indicating active support efforts. However, 34 open tickets represent a significant backlog that may negatively impact customer satisfaction.

Additionally, the presence of both “Resolved” and “Closed” suggests inconsistency in status classification, which may affect performance tracking.

Issue Type Analysis

The most common issues identified include:

- Bug Reports (most frequent)
- Service Delay
- Locked Account issues
- Payment-related issues

These trends indicate that system reliability and service efficiency are key drivers of customer complaints

Support Agent Performance

- John handled the highest number of tickets, making him the busiest agent
- Ticket distribution among agents is uneven

This imbalance may lead to inefficiencies and increased workload pressure on certain agents.

Department Analysis

The Billing department recorded the highest number of complaints.

This suggests potential issues in billing processes, transactions, or customer communication.

Customer Complaint Behavior

The analysis reveals repeated complaints and high-frequency customers:

- C115 and C121 reported Locked Account issues 3 times each
- C139 reported Bug Report issues 3 times.

High-volume customers include:

- C122 (8 tickets)
- C106 (7 tickets)

These patterns indicate unresolved issues and recurring service failures.

Customer Risk & Retention Analysis

A small group of customers contributes a large portion of total complaints, indicating a high-risk churn segment.

Customers such as C122 and C106 are at risk of dissatisfaction due to repeated issues.

If not addressed, this may lead to customer churn (loss of customers).

Priority & Resolution Insights

Low priority tickets make up the largest share of tickets. They also account for the highest total resolution time

The average resolution time of 34 hours indicates moderate efficiency, with room for improvement.

Support Structure Insight

Support agents are not assigned to specific departments but handle tickets across all areas.

While this provides flexibility, it may reduce specialization and slow down resolution of complex or recurring issues.

Root Cause Analysis & Resolution Strategy

Repeated complaints indicate that issues are not being fully resolved but temporarily addressed.

Identified Recurring Issues

- Locked Account issues (C115, C121)
- Bug Reports (C139)
- Service Delays (multiple customers)

Root Cause Identification

Using a structured approach:

Example:

- Repeated account lockouts
- Caused by failed login attempts
- Due to unclear password reset process
- Resulting from poor user interface design

Root cause: system usability issue

Resolution Strategy

To address root causes effectively:

- Fix recurring system bugs permanently
- Improve user experience (login, password reset)
- Automate repetitive tasks
- Monitor repeated complaints and act proactively

Expected Impact

- Reduced repeated complaints
- Improved efficiency
- Lower support workload
- Increased customer satisfaction

Key Insights

- Repeated complaints indicate unresolved root causes
- A small number of customers contribute significantly to total tickets
- Bug Reports and Service Delays are the most common issues
- Workload is uneven among support agents
- Billing is a major source of complaints
- Open tickets indicate backlog and potential delays
- Lack of specialization may reduce efficiency
- High-frequency customers represent churn risk

Recommendations

Address Recurring Issues

Identify patterns and fix root causes permanently

Proactively Engage High-Risk Customers

- Contact customers with multiple complaints (e.g., C122, C106)
- Provide personalized support

Improve System Reliability

Fix bugs and improve system stability

Optimize Agent Workload

- Distribute tickets evenly
- Monitor performance

Introduce Specialization

Assign agents to departments

Standardize Ticket Status

Clearly define “Resolved” vs “Closed”

Reduce Open Tickets

- Prioritize backlog clearance
- Implement SLAs(Service Legal Agreement)

Automate Low-Priority Tasks

Improve efficiency by automating repetitive issues

Conclusion

This analysis highlights key inefficiencies in customer support operations, particularly in recurring issues, workload imbalance, and resolution delays.

A small group of customers contributes a large portion of complaints, making proactive engagement and retention strategies essential.

By addressing root causes, improving system performance, and optimizing support processes, the organization can significantly enhance customer satisfaction, reduce complaints, and improve operational efficiency.

THANK YOU